Class 05: Data Visualization with GGPLOT

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Background

There are two ways to create visualizations in R: using base R functions or using the ggplot2 package. Data visualizations created with ggplot2 are often more aesthetically pleasing and easier to customize than those created with base R functions.

This is an example of making a scatter plot of the cars dataset using base R functions and ggplot2:

```
head(cars,3)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
```

Base R:

```
plot(cars)
```

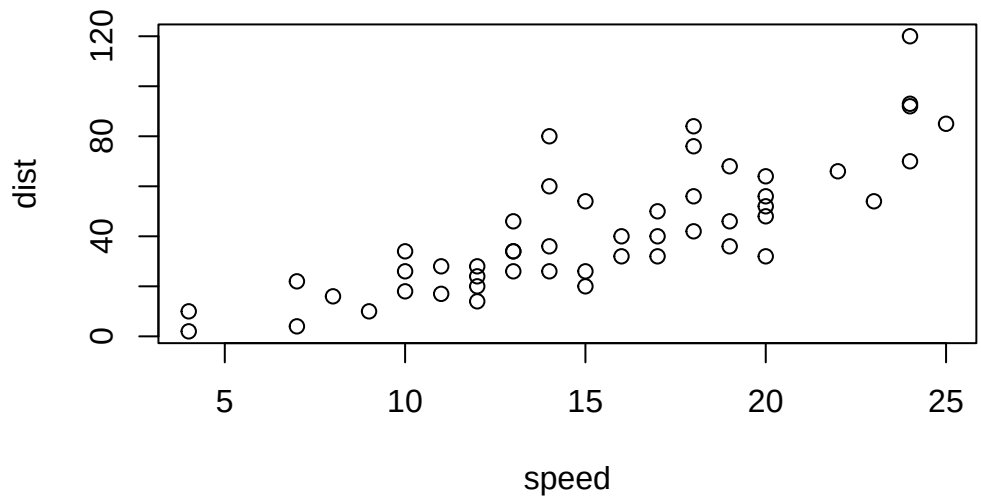

Figure 1: Base R plot of the cars dataset

ggplot2:

```
#install.packages("ggplot2")
library(ggplot2)
ggplot(cars)
```

Figure 2: example ggplot2 without setting dataset

Every ggplot needs atleast 3 things:

- A dataset
- Aesthetic mappings (aes)
- Geometric objects (geoms)

```
library(ggplot2)
ggplot(data = cars, aes(x = speed, y = dist)) +
  geom_point()
```

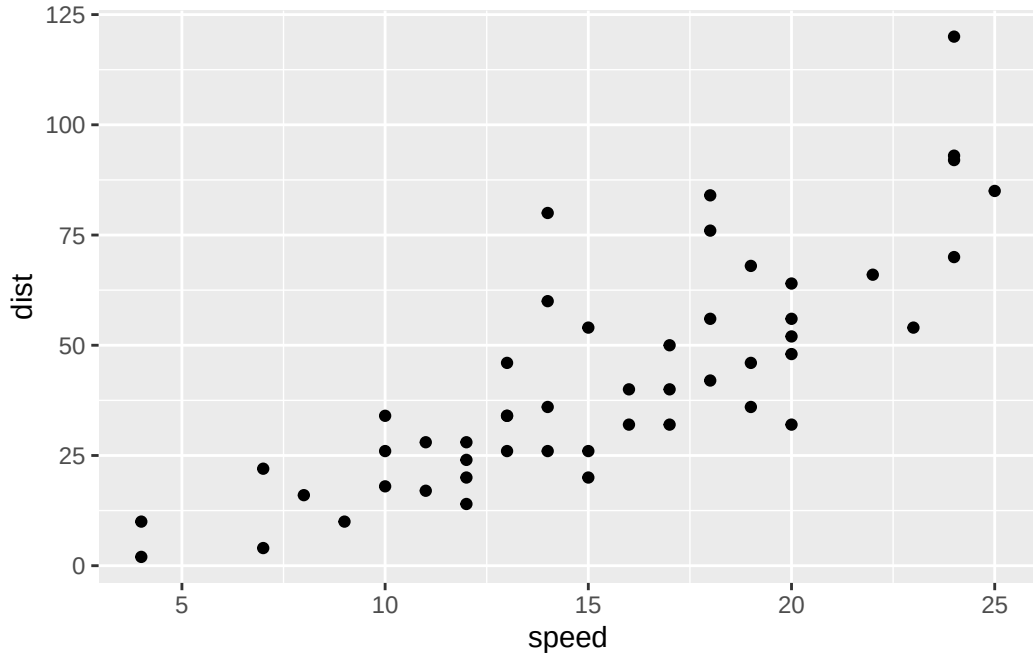

Figure 3: ggplot2 scatter plot example

Q1. For which phases is data visualization important in our scientific workflows?

Ans: Communication of Results, Exploratory Data Analysis (EDA), and Detection of outliers.

Q2. True of False? The ggplot2 package comes already installed with R?

Ans: False

Common Plot Types

Q. Which plot types are typically NOT used to compare distributions of numeric variables?

Ans: Network graphs

Q. Which statement about data visualization with ggplot2 is incorrect?

Ans: ggplot2 is the only way to create plots in R

Creating Scatter Plots

Q. Which geometric layer should be used to create scatter plots in ggplot2?

Ans: geom_point()

Note: These next questions refer to the ggplot scatter plot created from the cars dataset above.

Q. In your own RStudio can you add a trend line layer to help show the relationship between the plot variables with the geom_smooth() function?

```
ggplot(data = cars, aes(x = speed, y = dist)) +
  geom_point() +
  geom_smooth()
```

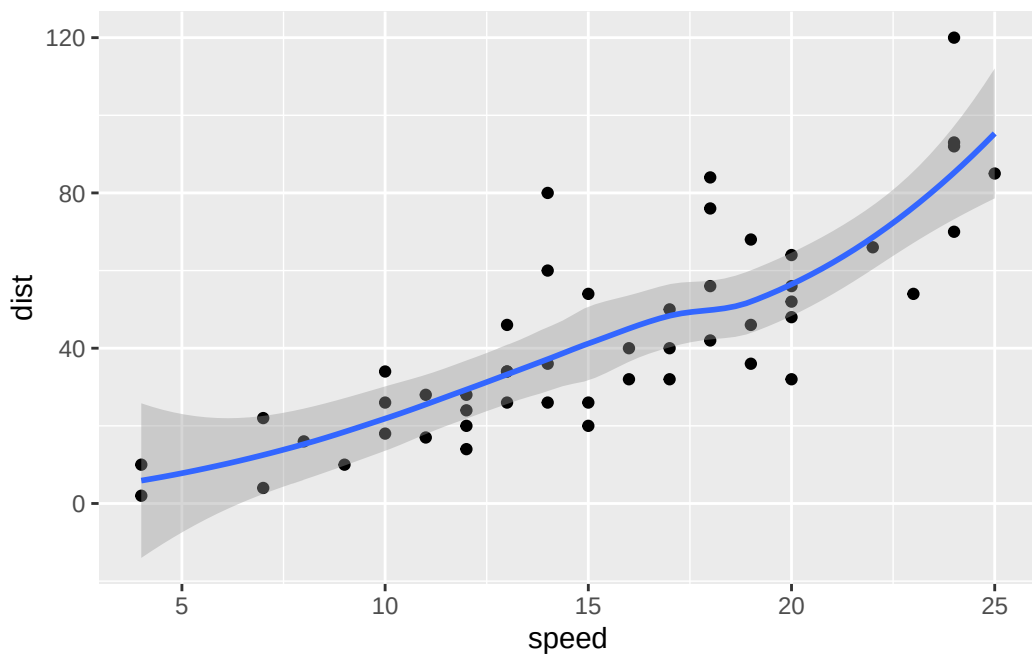

Figure 4: ggplot2 scatter plot with trend line

Q. Argue with geom_smooth() to add a straight line from a linear model without the shaded standard error region?

```
ggplot(data = cars, aes(x = speed, y = dist)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE)
```

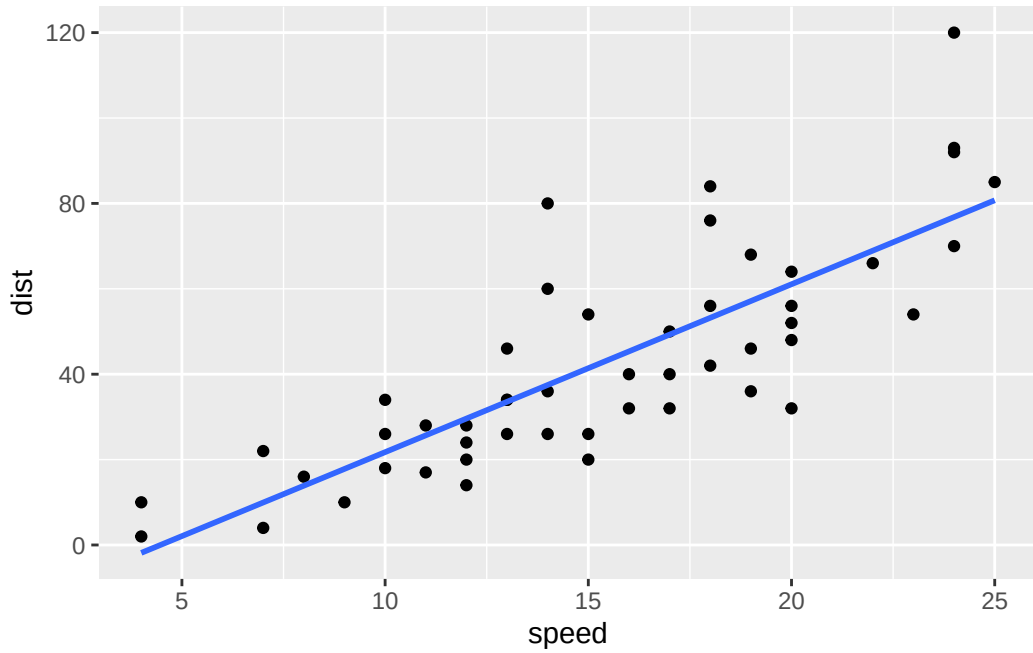

Figure 5: ggplot2 scatter plot with linear trend line

Q. Can you finish this plot by adding various label annotations with the labs() function and changing the plot look to a more conservative "black & white" theme by adding the theme_bw() function?

```
ggplot(data = cars, aes(x = speed, y = dist)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Stopping Distance vs. Speed of Cars",
       x = "Speed (mph)",
       y = "Stopping Distance (ft)") +
  theme_bw()
```

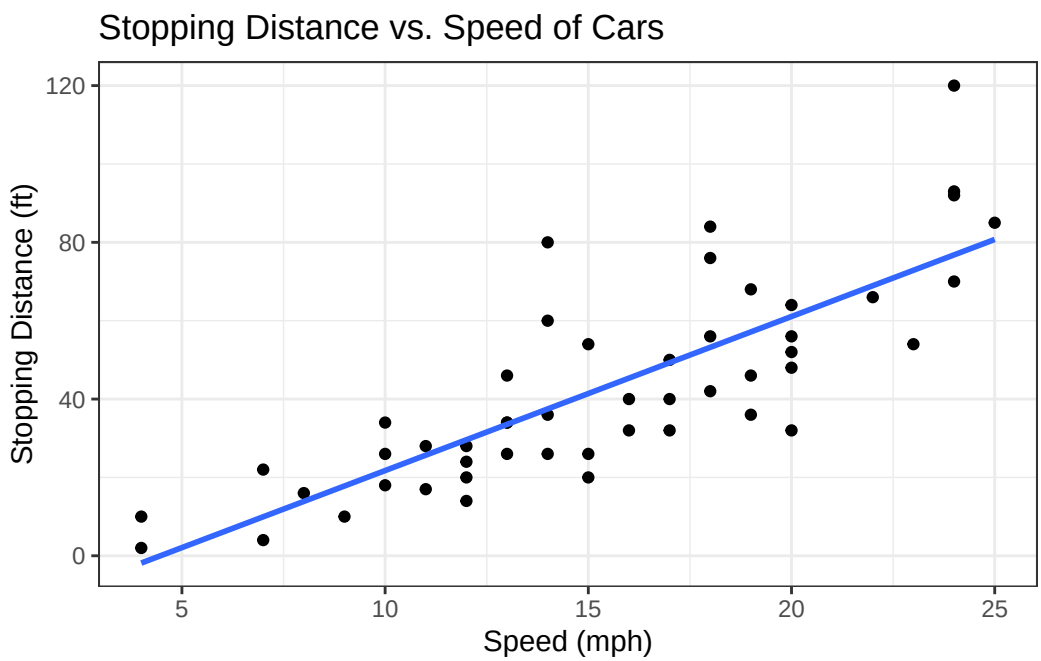

Figure 6: ggplot2 scatter plot with labels and black & white theme

Gene Expression Plot

Read the expression data from the class website:

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

> Q. Use the nrow() function to find out how many genes are in this dataset. What is your answer?

```r
nrow(genes)
```

```
[1] 5196
```

Q. Use the colnames() function and the ncol() function on the genes data frame to find out what the column names are (we will need these later) and how many columns there are. How many columns did you find?

```r
colnames(genes)
```

```
[1] "Gene"       "Condition1" "Condition2" "State"
```

```r
ncol(genes)
```

```
[1] 4
```

Q. Use the table() function on the State column of this data.frame to find out how many 'up' regulated genes there are. What is your answer?

```r
table(genes$State)
```

```
      down unchanging         up 
        72       4997        127 
```

Q. Using your values above and 2 significant figures. What fraction of total genes is up-regulated in this dataset?

```r
total_upreg <- table(genes$State)["up"]
total_genes <- nrow(genes)
fraction_upreg <- round(total_upreg / total_genes, 2)
fraction_upreg
```

```
  up 
0.02 
```

First Expression Scatter Plot

```r
ggplot(genes, aes(x = Condition1, y = Condition2)) +
  geom_point()
```

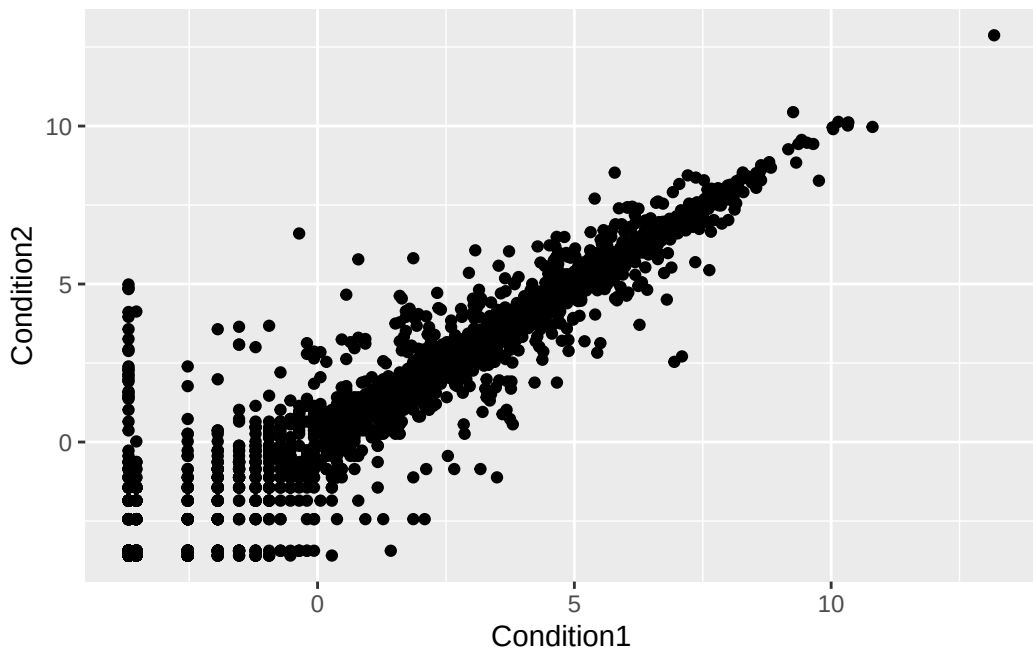

Figure 7: Gene expression scatter plot

Second Expression Scatter Plot

```r
ggplot(genes, aes(x = Condition1, y = Condition2, color = State)) +
  geom_point()
```

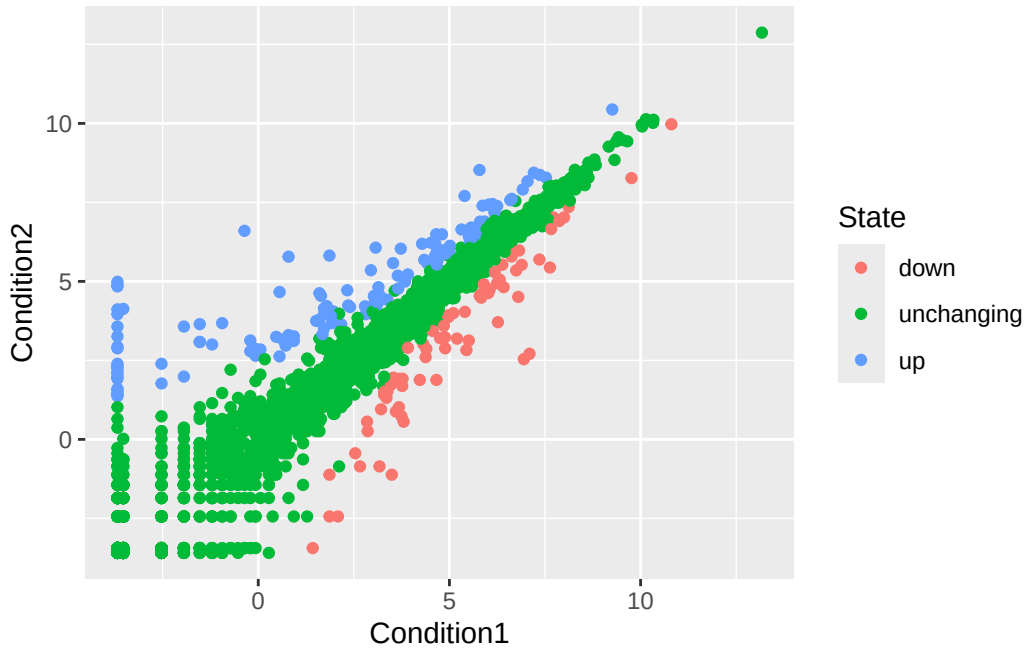

Figure 8: Gene expression scatter plot colored by State

Third Expression Scatter Plot

```
ggplot(genes, aes(x = Condition1, y = Condition2, color = State)) +
  geom_point() +
  scale_color_manual(values = c("blue", "gray", "red")) +
  labs(title = "Gene Expression Changes Upon Drug Treatment",
       x = "Control (no drug)",
       y = "Drug Treatment") +
  theme_bw()
```

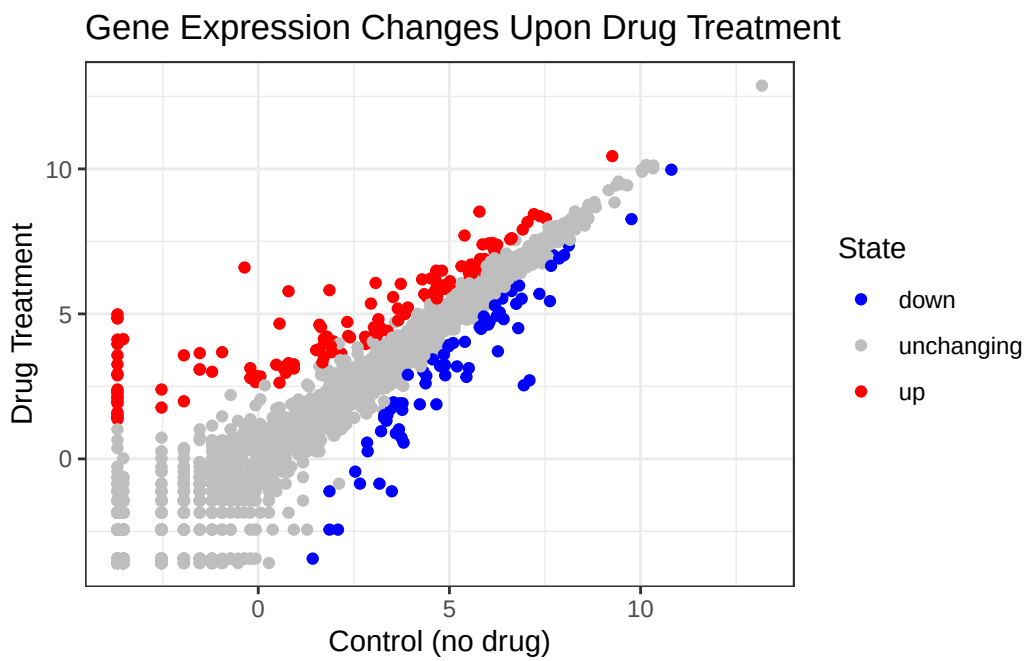

Figure 9: Gene expression scatter plot with labels and theme

Going Further

Read the gapminder dataset:

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv
gapminder <- read.delim(url)
head(gapminder)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
4 Afghanistan      Asia 1967  34.020 11537966  836.1971
5 Afghanistan      Asia 1972  36.088 13079460  739.9811
6 Afghanistan      Asia 1977  38.438 14880372  786.1134
```

Use dplyr to filter for the year 2007:

```r
library(dplyr)
gapminder2007 <- gapminder %>% filter(year == 2007)
head(gapminder2007)
```

```
      country continent year lifeExp      pop  gdpPercap
1 Afghanistan      Asia 2007  43.828 31889923   974.5803
2     Albania    Europe 2007  76.423  3600523  5937.0295
3     Algeria    Africa 2007  72.301 33333216  6223.3675
4      Angola    Africa 2007  42.731 12420476  4797.2313
5   Argentina  Americas 2007  75.320 40301927 12779.3796
6   Australia   Oceania 2007  81.235 20434176 34435.3674
```

Q. Complete the code below to produce a first basic scater plot of this gapminder_2007 dataset:

```r
ggplot(gapminder2007, aes(x = gdpPercap, y = lifeExp)) +
  geom_point()
```

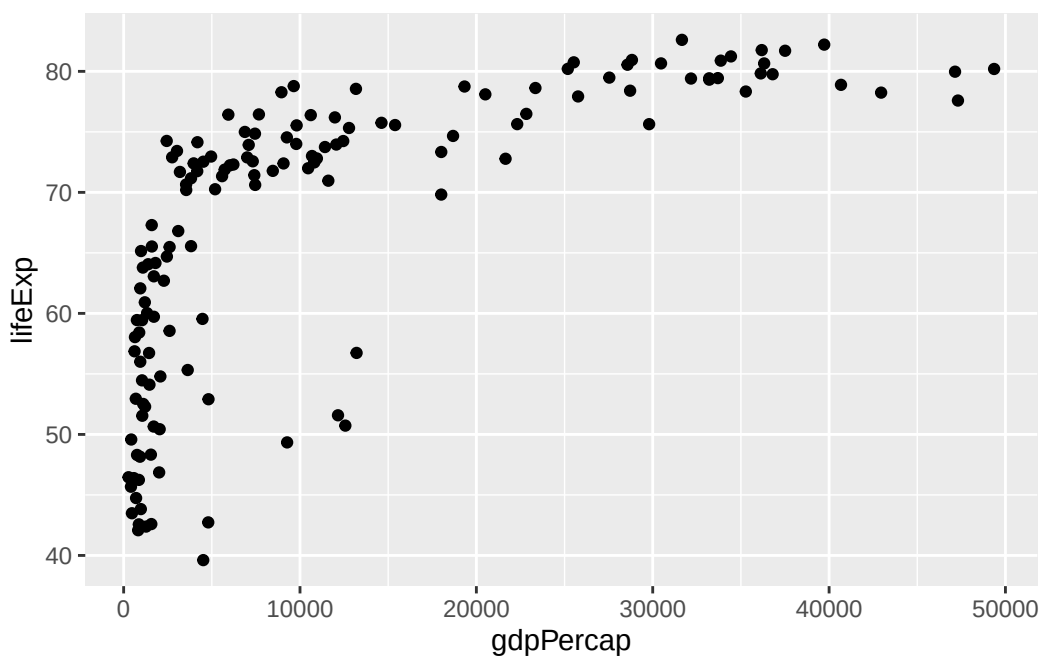

Figure 10: Gapminder 2007 scatter plot of GDP per capita vs Life Expectancy

Q. Can you adapt the code you have learned thus far to reproduce our gapminder scatter plot for the year 1957? What do you notice about this plot is it easy to compare with the one for 2007?

```
gapminder_1957 <- gapminder %>% filter(year == 1957)
ggplot(gapminder_1957, aes(x = gdpPercap, y = lifeExp)) +
  geom_point(aes(color = continent, size = pop), alpha = 0.7) +
  scale_size_area(max_size = 15)
```

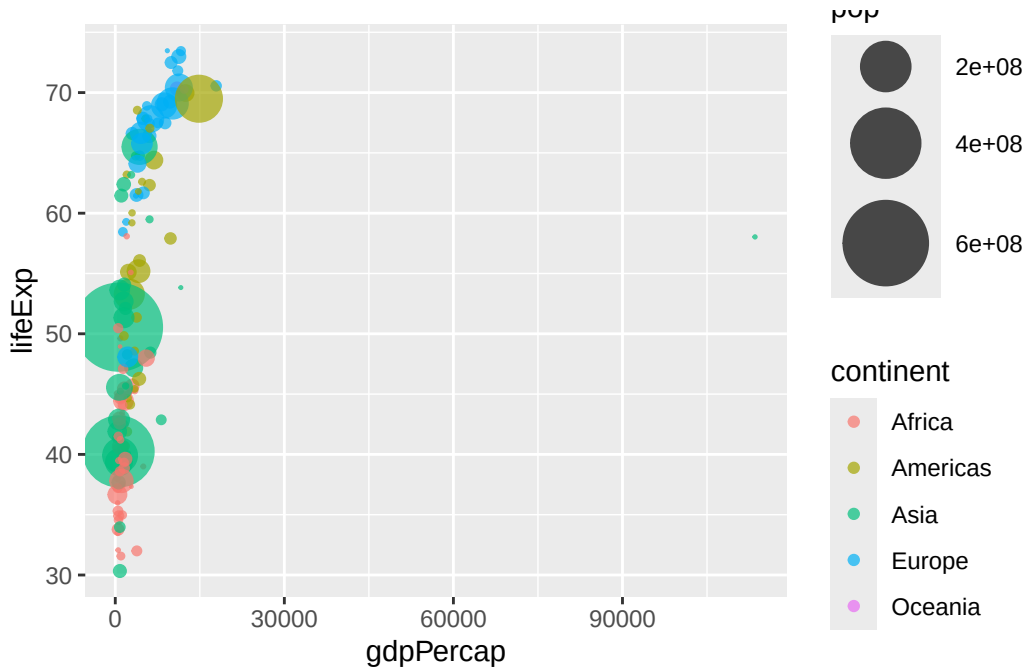

Figure 11: Gapminder scatter plot for 1957

Q. Do the same steps above but include 1957 and 2007 in your input dataset for ggplot(). You should now include the layer facet_wrap(~year) to produce the following plot:

```
gapminder_5707 <- gapminder %>% filter(year == 1957 | year == 2007)
ggplot(gapminder_5707, aes(x = gdpPercap, y = lifeExp)) +
  geom_point(aes(color = continent, size = pop), alpha = 0.7) +
  scale_size_area(max_size = 8) +
  facet_wrap(~year)
```

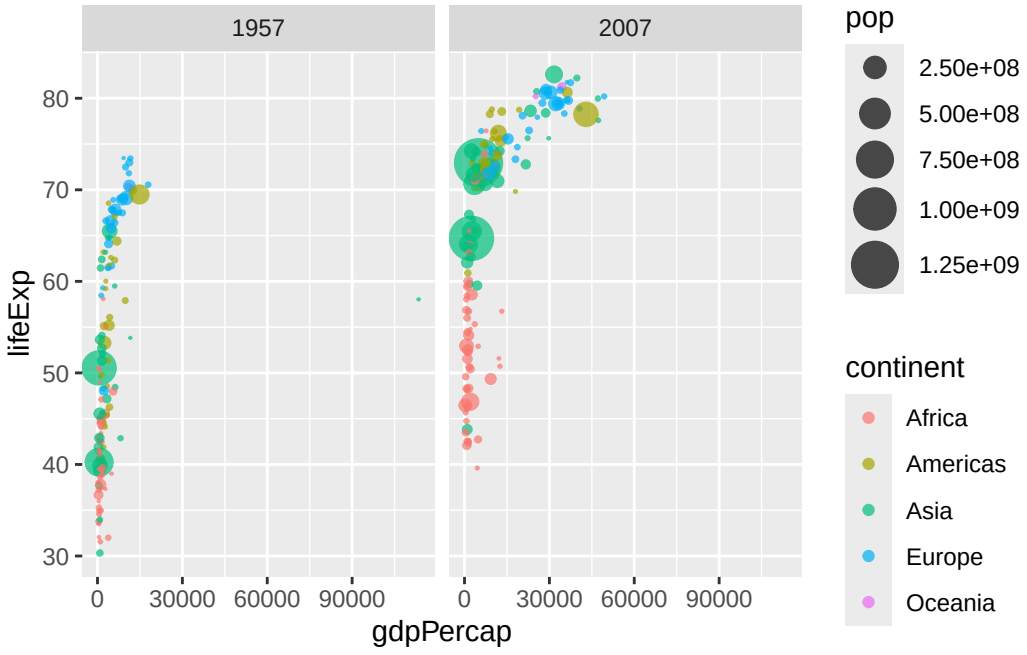

Figure 12: Gapminder scatter plot faceted by year